sunlight-market

December 22, 2025

0.1 Does solar activity affect the market?

This notebook investigates whether the certain astronomical feature (sunspot frequency) provides meaningful information for the behavior of the S&P 500 log returns.

0.1.1 TL;DR

Our analysis, based on graphical inspection and linear OLS-based regression models, found no statistical evidence that the monthly average sunspot frequency (as measured by SILSO) significantly influences the daily returns of the S&P 500. As such, we would not recommend using such an exogeneous variable for informing investment decisions.

0.2 Instructions for running the notebook

Install uv on your machine. Mac users can simply run

```
brew install uv
```

Then, execute

```
uv init
source .venv/bin/activate
uv add ipykernel
uv add matplotlib
uv add pandas
uv add yfinance
uv add statsmodels
uv add scikit-learn
```

This will allow you to run the cells below.

1 Loading packages

[1]:
```
import os
import yfinance

import numpy as np
import pandas as pd

import matplotlib.pyplot as plt
```

```
import statsmodels.api as sm

from statsmodels.graphics.tsaplots import plot_acf
from datetime import datetime
from dateutil import relativedelta
```

2 Loading data

[2]:
```
# We will get the daily data from the last 20 years (2005-2025)
START_DATE = datetime(2005, 9, 30)
DELTA_YEARS = relativedelta.relativedelta(years=20)

raw_data = yfinance.download(
    tickers="^GSPC", start=START_DATE, end=START_DATE + DELTA_YEARS,␣
 ↪interval="1d"
)

# As we are simply analzing the S&P 500 index, we can extract its data from the␣
 ↪multi-index dataframe
raw_data = raw_data.xs("^GSPC", level="Ticker", axis=1)
raw_data.head()
```

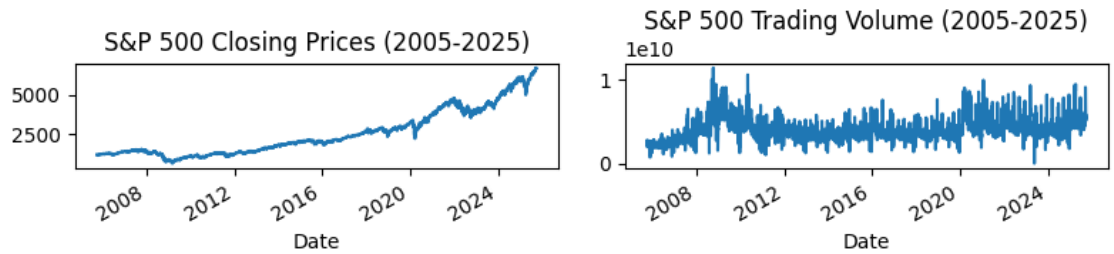
```
/var/folders/29/xv4b56h92vn6yl3hd0k5btw80000gn/T/ipykernel_46255/3500756974.py:5
: FutureWarning: YF.download() has changed argument auto_adjust default to True
  raw_data = yfinance.download(
[*********************100%***********************]  1 of 1 completed
```

[2]:

Price	Close	High	Low	Open	Volume
Date					
2005-09-30	1228.810059	1229.569946	1225.219971	1227.680054	2097520000
2005-10-03	1226.699951	1233.339966	1225.150024	1228.810059	2097490000
2005-10-04	1214.469971	1229.880005	1214.020020	1226.699951	2341420000
2005-10-05	1196.390015	1214.469971	1196.250000	1214.469971	2546780000
2005-10-06	1191.489990	1202.140015	1181.920044	1196.390015	2792030000

2.1 Exploratory Analyis

[3]:
```
plt.figure(figsize=(8, 2))
plt.subplot(1, 2, 1)
raw_data.Close.plot(title="S&P 500 Closing Prices (2005-2025)")
plt.subplot(1, 2, 2)
raw_data.Volume.plot(title="S&P 500 Trading Volume (2005-2025)")
plt.tight_layout()
```



Under a standard geometric Brownian motion model, log-returns are normally distributed; therefore, we will consider henceforth the effect of the solar activity on the daily closing prices log-returns.

[4]:
```
data = raw_data.copy()
data["LogReturn"] = (data["Close"] / data["Close"].shift(1)).apply(lambda x: np.
 ↪log(x))
# We drop the first row as it will have NaN log-return
data = data.iloc[1:, :]

# We also include both the Month and Year columns for merging purposes (see␣
 ↪below)
data["Year"] = data.index.year
data["Month"] = data.index.month
data["Day"] = data.index.day
assert not data["LogReturn"].isna().any()

data = data[["LogReturn", "Day", "Year", "Month"]]
```

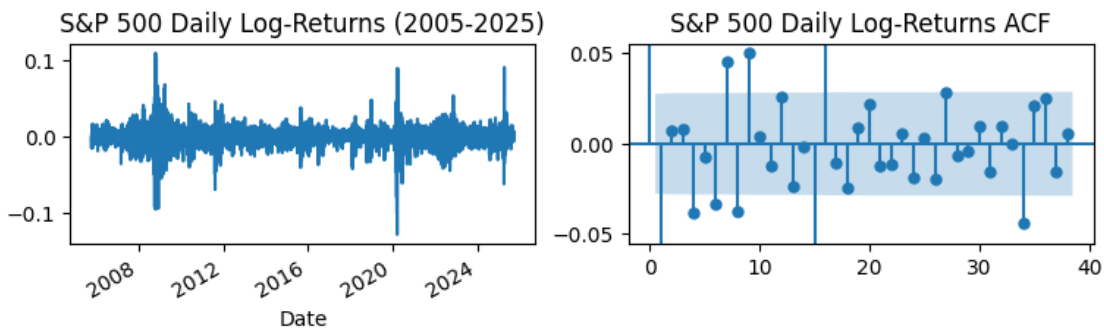
[5]:
```
plt.figure(figsize=(9, 2))
ax1 = plt.subplot(1, 2, 1)
data.LogReturn.plot(title="S&P 500 Daily Log-Returns (2005-2025)", ax=ax1)
ax2 = plt.subplot(1, 2, 2)

# This is the expected autocorrelation from a white noise process
ci = 1.96 / np.sqrt(len(data))

# We plot the ACF for the log-returns
plot_acf(
    data.LogReturn,
    ax=ax2,
    title="S&P 500 Daily Log-Returns ACF",
    alpha=0.05,
    missing="raise",
)

# To clearly see the confidence intervals, we limit the y-axis
ax2.set_ylim(-2 * ci, 2 * ci)
```

```
plt.show()
```



This plot suggests that there is no strong temporal correlation between the S&P 500's log returns (which is a known general fact about financial time series).

In fact, financial data has been widely studied, and stylized facts regarding their behavior are well known, see, e.g., (McNeil et al., 2005, Chapters 3-4). Nonetheless, a comprehensive analysis with sophisticated volatility models (e.g., GARCH) is beyond the scope of this notebook. We will focus on a simple OLS-based regression for assessing the relationship between sunspot frequency and the S&P500 returns.

Towards this objective, we will first fetch the needed information on our chosen exogeneous variable from SILSO.

[6]:
```
filename = "sunspots.csv"
if not os.path.exists(filename):
    !curl -L https://www.sidc.be/SILSO/INFO/snmtotcsv.php -o $filename
```

[7]:
```
# Column info available at https://www.sidc.be/SILSO/infosnmtot.
sunspots = pd.read_csv(
    "sunspots.csv",
    names=[
        "Year",
        "Month",
        "DateFrac",
        "MonthlyMean",
        "MonthlyStdDev",
        "Observations",
        "IsDefinitive",
    ],
    delimiter=";",
)

# We will only need the data after 2005-12
sunspots = sunspots[
```

```
    (sunspots["Year"] > START_DATE.year) | ((sunspots["Year"] == START_DATE.
 ↪year) & (sunspots["Month"] >= START_DATE.month))
]
# We create a date column for merging purposes
sunspots["Date"] = pd.to_datetime(
    sunspots["Year"].astype(str) + "-" + sunspots["Month"].astype(str)
)
```

[8]:
```
# Verify whether some months are not covered
def check_missing_months(sunspots):
    dates = pd.to_datetime(dict(year=sunspots.Year, month=sunspots.Month,␣
 ↪day=1))

    full = pd.date_range(dates.min(), dates.max(), freq="MS")

    missing = full.difference(dates)
    return missing


missing_months = check_missing_months(sunspots)
assert len(missing_months) == 0, f"Missing months: {missing_months}"
# Check for missing values
assert not sunspots.MonthlyMean.isna().any()

sunspots.head()
```

[8]:
```
      Year  Month  DateFrac  MonthlyMean  MonthlyStdDev  Observations  \
3080  2005      9  2005.707         37.2            4.5           691
3081  2005     10  2005.790         13.2            3.4           534
3082  2005     11  2005.874         27.5            4.1           499
3083  2005     12  2005.958         59.3            3.9           419
3084  2006      1  2006.042         20.9            3.4           511

      IsDefinitive       Date
3080             1 2005-09-01
3081             1 2005-10-01
3082             1 2005-11-01
3083             1 2005-12-01
3084             1 2006-01-01
```

We then merge the data sets based on the yearly and monthly data. As our `sunspots` data set contains averages from the entire month, we avoid data snooping by only using data from the previous month at each day.

[10]:
```
sunspots["NextMonth"] = sunspots["Month"] + 1
sunspots["NextMonth"] = sunspots["NextMonth"].replace(13, 1)
```

```
sunspots["NextYear"] = (sunspots["Month"] == 12).astype(int) + sunspots["Year"]

# We drop the last row as it will have NaN in NextMonth
sunspots = sunspots.iloc[:-1, :]

sp500_sunspots = data.merge(
    sunspots[["NextYear", "NextMonth", "MonthlyMean"]],
    left_on=["Year", "Month"],
    right_on=["NextYear", "NextMonth"],
    how="left",
    suffixes=("", "_Sunspots"),
)

# We drop the NextMonth column (which was only used for merging)
sp500_sunspots = sp500_sunspots.drop(columns=["NextMonth"])
```

[13]:
```
assert not sp500_sunspots["MonthlyMean"].isna().any()
sp500_sunspots.head()
```

[13]:

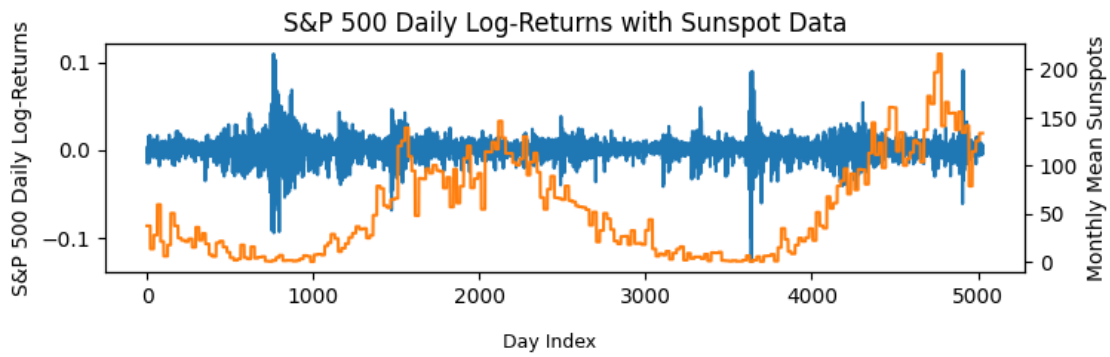
	LogReturn	Day	Year	Month	NextYear	MonthlyMean
0	-0.001719	3	2005	10	2005	37.2
1	-0.010020	4	2005	10	2005	37.2
2	-0.014999	5	2005	10	2005	37.2
3	-0.004104	6	2005	10	2005	37.2
4	0.003694	7	2005	10	2005	37.2

For exploration, we plot both time series.

[14]:
```
fig = plt.figure(figsize=(8, 2))
ax1 = sp500_sunspots.LogReturn.plot(title="S&P 500 Daily Log-Returns with␣
 ↪Sunspot Data")
ax2 = sp500_sunspots.MonthlyMean.plot(
    secondary_y=True,
    label="Monthly Mean Sunspots",
)

ax1.set_ylabel("S&P 500 Daily Log-Returns")
ax2.set_ylabel("Monthly Mean Sunspots")
fig.supxlabel("Day Index", y=-0.16, fontsize=9)

plt.show()
```



We are now able to assess whether the average number of monthly sunspots significantly influences the daily log-returns for the S&P00.

2.2 Time Series Modelling

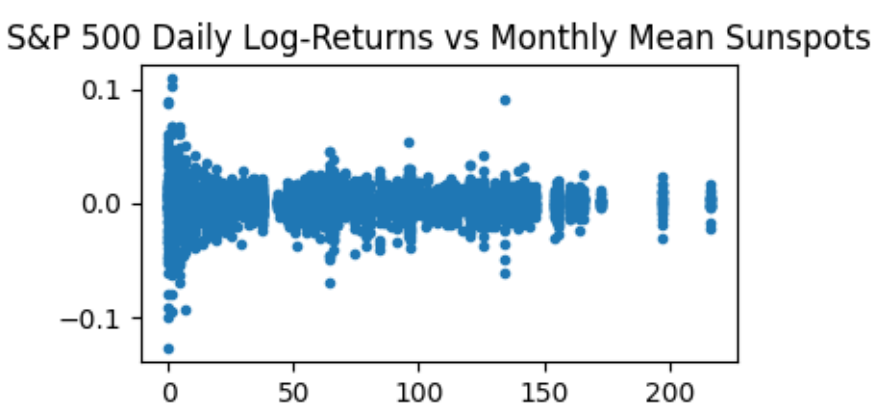
We provide a simple, albeit recognizably competitive, model for our time series. Specifically, let $\mathbf{x} = \{x_t\}_{t=1}^T$ and $\mathbf{y} = \{y_t\}_{t=1}^T$ be the SILSO-measured sunspot frequency and the S&P 500 time series, respectively.

We will start, however, by measuring the (cross-) correlation between $\mathbf{x}$ and $\mathbf{y}$. We also plot both $\mathbf{x}$ and $\mathbf{y}$ in a scatter plot to search for any obvious tendencies (or lack thereof).

[30]:
```
raw_x = sp500_sunspots["MonthlyMean"]
raw_y = sp500_sunspots["LogReturn"]

plt.figure(figsize=(4, 2))
plt.plot(raw_x, raw_y, marker=".", linestyle="none")
plt.title("S&P 500 Daily Log-Returns vs Monthly Mean Sunspots")
```

[30]:
```
Text(0.5, 1.0, 'S&P 500 Daily Log-Returns vs Monthly Mean Sunspots')
```



There is no clear sign that the sunspot frequency has any effect on the S&P 500 index, indeed, the conditional distribution $\mathbb{P}[Y_t|X_t]$ looks symmetric to the naked eye. To validate this, we can simply test whether $\beta = 0$ in the following model.

$$y_t \sim \mathcal{N}(\beta x_t, \sigma^2),$$

with $\mathcal{N}$ as the Gaussian distribution and β as an unknown coefficient.

[31]:
```
ols = sm.OLS(raw_y, raw_x).fit()
ols.summary()
```

[31]:

Dep. Variable:	LogReturn	**R-squared (uncentered):**	0.001
Model:	OLS	**Adj. R-squared (uncentered):**	0.001
Method:	Least Squares	**F-statistic:**	4.036
Date:	Mon, 22 Dec 2025	**Prob (F-statistic):**	0.0446
Time:	11:51:16	**Log-Likelihood:**	14986.
No. Observations:	5029	**AIC:**	-2.997e+04
Df Residuals:	5028	**BIC:**	-2.996e+04
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
MonthlyMean	4.757e-06	2.37e-06	2.009	0.045	1.15e-07	9.4e-06

Omnibus:	1170.595	**Durbin-Watson:**	2.242
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	34867.058
Skew:	-0.454	**Prob(JB):**	0.00
Kurtosis:	15.867	**Cond. No.**	1.00

Notes:
[1] R^2 is computed without centering (uncentered) since the model does not contain a constant.
[2] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Two properties stand out from this summary.

1. The sunspot frequency alone does not provide a sufficient explanation for the variance of the S&P 500 (which is obvious and expected).
2. Although there is enough evidence (according to a t-test) to reject the null hypothesis that $\beta = 0$, the coefficient is practically significant.

2.3 Regression models

We will also consider a simple regression model, defined by

$$y_t \sim \mathcal{N}(\beta_1 x_t + \beta_2 y_{t-1} + \beta_3, \sigma^2),$$

in which β_1, β_2 are to-be-estimated coefficients; σ^2 is fixed but unknown. As x_t represents the monthly average number of sunspots at the $t-1$ month, we can safely assume that such a measurement will be available at the t-th stage. This is a simple autoregressive (AR) model with exogeneous inputs (ARX).

[44]:
```python
def ols_one_step_ahead_lagged(raw_x, raw_y):

    y = raw_y[1:]
    X = pd.DataFrame({
        "Sunspots": raw_x[1:],
        "LaggedLogReturn": raw_y[:-1].values,
    })

    # Adding a constant is equivalent to estimating $\beta_{3}$ (the intercept)
    X = sm.add_constant(X)

    model = sm.OLS(y, X).fit()
    return model

model = ols_one_step_ahead_lagged(raw_x, raw_y)
model.summary()
```

[44]:

Dep. Variable:	LogReturn_Causal	**R-squared:**	0.006
Model:	OLS	**Adj. R-squared:**	0.006
Method:	Least Squares	**F-statistic:**	14.98
Date:	Mon, 22 Dec 2025	**Prob (F-statistic):**	3.25e-07
Time:	12:19:33	**Log-Likelihood:**	15617.
No. Observations:	5028	**AIC:**	-3.123e+04
Df Residuals:	5025	**BIC:**	-3.121e+04
Df Model:	2		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	0.0002	0.000	0.943	0.346	-0.000	0.001
Sunspots	2.868e-06	3.1e-06	0.927	0.354	-3.2e-06	8.94e-06
LaggedLogReturn	-0.0760	0.014	-5.405	0.000	-0.104	-0.048

Omnibus:	535.366	**Durbin-Watson:**	2.000
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	2029.510
Skew:	-0.490	**Prob(JB):**	0.00
Kurtosis:	5.954	**Cond. No.**	6.74e+03

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 6.74e+03. This might indicate that there are
strong multicollinearity or other numerical problems.

We again highlight two notable properties from the above results.

1. Conditionally on the lagged log-return, the monthly average sunspot frequency does not significantly influence the current log-returns.
2. Our data does not sufficiently explain the variability of the observed log returns ($R^2 \approx 10^{-2}$).

2.4 Dealing with non-Gaussianity

Obviously, financial data is massively non-Gaussian due to their fat tails, and the above models can only provide educated (albeit arguably reasonable) guesses. To mitigate this issue, we winsorize the data.

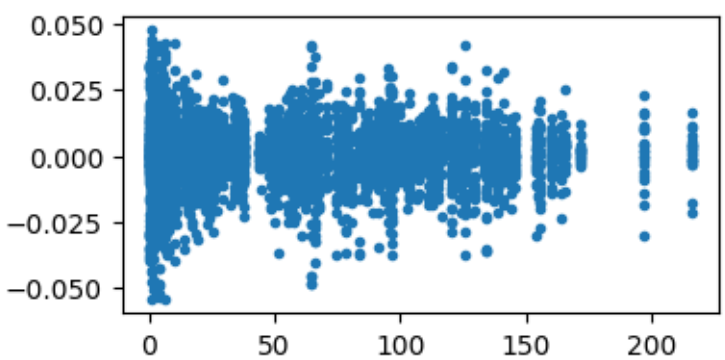
[56]:
```
# We winsorize the data to mitigate the lack of Gaussianity.
# We also use a causal approach, i.e., we only use past data to compute the␣
↪bounds.
lower_bound = sp500_sunspots['LogReturn'].expanding().quantile(0.01)
upper_bound = sp500_sunspots['LogReturn'].expanding().quantile(0.99)

# Clip the values using the bounds calculated from past data
sp500_sunspots['LogReturn_Causal'] = sp500_sunspots['LogReturn'].
↪clip(lower=lower_bound, upper=upper_bound)

raw_x = sp500_sunspots["MonthlyMean"]
raw_y = sp500_sunspots["LogReturn_Causal"]

plt.figure(figsize=(4, 2))
plt.plot(raw_x, raw_y, marker=".", linestyle="none")
```

[56]: [<matplotlib.lines.Line2D at 0x1506f0910>]



[57]:
```
model = ols_one_step_ahead_lagged(raw_x, raw_y)
model.summary()
```

[57]:

Dep. Variable:	LogReturn_Causal	**R-squared:**	0.006
Model:	OLS	**Adj. R-squared:**	0.006
Method:	Least Squares	**F-statistic:**	14.98
Date:	Mon, 22 Dec 2025	**Prob (F-statistic):**	3.25e-07
Time:	12:30:44	**Log-Likelihood:**	15617.
No. Observations:	5028	**AIC:**	-3.123e+04
Df Residuals:	5025	**BIC:**	-3.121e+04
Df Model:	2		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	0.0002	0.000	0.943	0.346	-0.000	0.001
Sunspots	2.868e-06	3.1e-06	0.927	0.354	-3.2e-06	8.94e-06
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Omnibus:	535.366	**Durbin-Watson:**	2.000
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	2029.510
Skew:	-0.490	**Prob(JB):**	0.00
Kurtosis:	5.954	**Cond. No.**	6.74e+03

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 6.74e+03. This might indicate that there are
strong multicollinearity or other numerical problems.

Importantly, our conclusions remain the same when outliers are clipped; sunspot frequency has no significant influence on log-returns.

2.5 Conclusions

We found no significant and exploitable statistical correlation between the monthly average sunspot frequency and the log-returns from the S&P 500. Although we have restricted our analysis to simple models—with constant volality—and a straightforward grahpical analysis, it remains unclear whether more sophisticated approaches would provide convincing evidence in favor of the influence of the astronomical factors on the market.

From an investment perspective, we believe that the monthly sunspot frequency *cannot* not be reliably used for informed decision-making. We leave a comprehensive investigation of such factors, however, to future work.